

AgentBoard: An Analytical Evaluation Board of Multi-turn LLM Agents

We introduce AgentBoard, and we show our benchmark and evaluation framework for multi-turn LLM agents.

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CODE & DATA

github.com/hkust-nlp/AgentBoard

SCOPE

Our analytical benchmark for LLM agents

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Abstract

- We present our unified benchmark and open-source evaluation framework for multi-turn LLM agents.
- We support partially-observable environments with multi-round interactions across diverse tasks.
- We introduce a fine-grained progress rate that captures incremental advancements.
- We enable multi-faceted analysis beyond success rate to improve interpretability.
- We aim to demystify agent behaviors and accelerate stronger agent development.

Our AgentBoard Overview



Figure: Overview of AGENTBOARD showing 9 diverse tasks and 1013 environments with multi-round, partially-observable interactions, plus the analytical evaluation toolkit.

- We cover 9 tasks and 1013 environments with human-verified subgoals.
- We provide our open-source toolkit with interactive visualizations (progress, sub-skills, grounding, long-range).

Our Principles & Contributions

- **Task Diversity & Uniformity** We curate planning-heavy, text-based, partially-observable tasks across embodied, game, web, and tool scenarios.
- **Fine-grained Progress Rate** We track incremental achievements via ordered, human-annotated subgoals.
- **Comprehensive Analysis** We include our grounding, difficulty breakdown, sub-skills, long-range behavior, and trajectories.

Our Unified Multi-round Agent

- We run a simple reflex agent that iteratively observes history and produces the next action toward the goal.

 Reflex agent loop schematic and example prompt: observe → decide → act across multi-round interactions

Figure: Reflex-loop architecture and prompt example illustrating iterative observe-decide-act with feedback in partially observable settings.

Our Progress Metric & Validation

- We decompose goals into subgoals and compute progress as the highest achieved match.
- We validate with a user study: 60 trajectories × 8 tasks × 3 models; 4 raters each.
- We observe high Pearson correlation (> 0.95) with human assessments and substantial Fleiss' κ .

 Correlation (>0.95) and Fleiss' kappa agreement between human progress ratings and automatic progress metric across tasks

Figure: Human vs. automatic progress—Pearson correlation exceeds 0.95 across tasks with substantial inter-annotator agreement (Fleiss' κ).

Our Environments

- Embodied: AlfWorld (subgoal-based progress + original success); ScienceWorld (re-annotated subgoals and guidance); BabyAI (textualized IO, denser subgoals).
- Game: Jericho (shortened goals to ~15 subgoals); PDDL (Gripper, Barman, Blocksworld, Tyreworld; text observations; matching-based progress).
- Web: WebShop (improved error feedback; distance-to-goal progress); WebArena (fused URL + content matching for continuous progress).
- Tool: Tool-Query (Weather/Movie/Academia; planned DB queries; subgoals); Tool-Operation (Todo & Sheets; subgoal or table-match progress; answers require valid tool trajectories).

Qualitative Trajectories (ALF · JC · WA · TO)

- ALF Goal: put a clean egg in microwave. Step 02: open fridge 1 → sees egg → Progress 0.00 → 0.25; then take egg → clean → microwave → finish.
- JC Goal: get out, then escape via driving. Step 29: take a shower → continue quickly → Progress 0.43 → 0.57.
- WA Goal: list issues in kkroening/ffmpeg-python with question-related labels. Step 05: click [5398] → Issues page focused → Progress 0.25 → 0.50.
- TO Goal: in Sheet17 compute Profit and sort desc. Step 07: update_cell_by_formula PRODUCT C8:D8 → E8 → Progress 0.47 → 0.49; then fill column → sort → finish.

Our Results & Analytics (Collapsed)

- We show that progress rate better differentiates models than success; proprietary models lead (GPT-4 highest overall).
- We observe advantages for code-pretrained and agent-tuned open-weight models.
- We analyze our grounding accuracy, difficulty breakdown, and long-range interaction; we find strong models sustain progress longer.

Backup: References & Resources

- Code & data: github.com/hkust-nlp/AgentBoard · Contact: llmagentboard@gmail.com
- Selected refs: Russell & Norvig (2005); OpenAI (2023); Touvron et al. (2023); Yao et al. (2023); Zhou et al. (2023); Shridhar et al. (2021); Chevalier-Boisvert et al. (2019); Wang et al. (2022); Vallati et al. (2015); Silver & Chitnis (2020); Yao et al. (2022)